

Matthew Runyan

📍 Boston, MA, USA

✉ runyan.m@northeastern.edu

📞 (912) 409-7654

🌐 mrunyan1

🔗 mrunyan1

Education

MS	Northeastern University , Bioinformatics Relevant coursework: Machine Learning & Data Mining; Genomics in Bioinformatics; Statistics for Bioinformatics.	Boston, MA, USA Sept 2023 – Apr 2025
BS	University of Virginia, School of Engineering and Applied Science , Biomedical Engineering Capstone project: Automated selection and placement of multicellular aggregates for bio-fabricated tissue constructs.	Charlottesville, VA, USA Aug 2018 – May 2022

Research Experience

Institute for Experiential AI, Northeastern University , ML Engineer, Computational Biology	Boston, MA, USA July 2024 – present
Develop deep learning models and genomic pipelines for pre-mRNA splicing and variant interpretation. - Designed and maintain a Nextflow-based pipeline to process RNA-seq and whole-genome sequencing samples from GTEx, ENCODE, and internal cohorts. - Develop dilated convolutional neural networks, masked language models, and transformer architectures for splice-site classification and splice-site usage prediction using personal transcriptomes. - Built a modular training framework with automated experiment tracking, hyperparameter logging, and evaluation reports to support large-scale model development. - Implemented model interpretability workflows using DeepLIFT/SHAP, seqlet discovery, and RBP motif analysis to identify cis-regulatory features driving predictions. - Contributed to a manuscript on personal-genome-trained splicing models, currently in preparation.	
Child Health Research Center, University of Virginia , Research Assistant	Charlottesville, VA, USA Feb 2022 – Aug 2023
Studied genetic and molecular mechanisms of pediatric kidney disease in Patricio Ray's laboratory. - Developed and maintained transgenic mouse and Drosophila lines for APOL1 kidney disease studies; performed animal husbandry and PCR/qPCR genotyping. - Performed molecular and cellular assays, including mammalian cell culture, histology, and immunohistochemistry, to characterize renal phenotypes. - Built and ran RNA-seq analysis pipelines for urine-derived renal epithelial cells, including QC, alignment, quantification, and differential expression. - Analyzed bulk RNA-seq data from multiple culture conditions to identify pathways and cell-type-specific markers associated with nephrotic syndrome. - Investigated cellular reprogramming conditions to expand urine-derived renal cells and characterized resulting populations using transcriptomic and imaging data.	

Industry Experience

NanoDx , Biomedical Engineer Intern	Southborough, MA, USA May 2021 – Aug 2021
Internship focused on wafer-level biosensor testing and device prototyping. - Used microfluidic and electronic probing to evaluate wafer-level semiconductor performance for biosensor production. - Designed and 3D-printed custom fixtures and mounts for probe positioning using CAD tools. - Communicated experimental results and design iterations to a multidisciplinary team of engineers.	

Projects

Automated selection and placement of multicellular aggregates	University of Virginia
--	------------------------

Biomedical Engineering capstone project developing an automated system for handling multicellular aggregates in biofabricated tissue constructs.

- Implemented MATLAB-based real-time image processing to detect and select multicellular aggregates.
- Integrated Arduino-controlled motors for precision placement and manipulation of aggregates.

Skills

Programming & Data Analysis: Python, R, MATLAB, SQL, Bash; Git, GitHub

Machine Learning & Statistics: TensorFlow, PyTorch, scikit-learn; deep learning for genomics; statistical modeling

Bioinformatics & Genomics: RNA-seq and WGS analysis; STAR, HTSeq, rMATS, LeafCutter, SpliSER, Nextflow, SAMtools, DESeq2

Laboratory Techniques: Mammalian cell culture, RNA/DNA extraction, PCR/qPCR, immunohistochemistry, mouse handling and genotyping

Computing & Platforms: Linux, SLURM/HPC, Docker, Conda, Jupyter, AWS

Publications

(manuscript in preparation) Deep learning models of alternative splicing with personal transcripts

Dec 2025

Matthew T. Runyan¹, Saumya Gupta¹, Jennifer Dy², Predrag Radivojac³, Peter Castaldi⁴, Ayan Paul^{1,4}

Affiliations: ¹The Institute for Experiential AI, Northeastern University; ²Department of Electrical and Computer Engineering, Northeastern University; ³Khoury College of Computer Sciences, Northeastern University; ⁴CDNM, Brigham & Women's Hospital, Harvard Medical School

Presentations

Conditional reprogramming of urinary epithelial cells cultured from children with nephrotic syndrome Pediatric Academic Societies Meeting, Washington, DC, USA

Poster presentation on conditional reprogramming and transcriptomic characterization of urine-derived epithelial cells from children with nephrotic syndrome.

Apr 2023

- Authors: Matthew T. Runyan, Jing Yu, Pankaj Kumar, and Patricio E. Ray.
- Affiliations: University of Virginia School of Medicine, Charlottesville, VA, USA.

Machine learning models for predicting variant effects in alternative splicing

ISMB 2025, Liverpool, UK

Poster presentation on deep learning models for variant effect prediction in alternative splicing.

July 2025

- Authors: Matthew T. Runyan¹, Saumya Gupta¹, Jennifer Dy², Predrag Radivojac³, Peter Castaldi⁴, and Ayan Paul^{1,4}.
- Affiliations: ¹The Institute for Experiential AI, Northeastern University; ²Department of Electrical and Computer Engineering, Northeastern University; ³Khoury College of Computer Sciences, Northeastern University; ⁴CDNM, Brigham & Women's Hospital, Harvard Medical School.

Machine learning models for predicting variant effects in alternative splicing

ASHG 2025, Boston, MA, USA

Poster presentation on deep learning models for variant effect prediction in alternative splicing.

Oct 2025

- Authors: Matthew T. Runyan¹, Saumya Gupta¹, Jennifer Dy², Predrag Radivojac³, Peter Castaldi⁴, and Ayan Paul^{1,4}.
- Affiliations: ¹The Institute for Experiential AI, Northeastern University; ²Department of Electrical and Computer Engineering, Northeastern University; ³Khoury College of Computer Sciences, Northeastern University; ⁴CDNM, Brigham & Women's Hospital, Harvard Medical School.